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On the Irregularity of the Image of the Iitaka Fibration[#]

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ABSTRACT

We characterize the irregularity of the image of the Iitaka fibration in terms of the dimension of certain cohomological support loci.

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1. INTRODUCTION

Let L be a Cartier divisor on a projective variety X with $\kappa(X, L) \geq 0$. The Iitaka fibration associated to L is a birational model of the map induced by the linear series $|mL|$ for m sufficiently big and divisible. We will denote it by $f_L: X' \rightarrow Y_L$, where X' is an appropriate birational model of X . It is a fundamental tool in the birational

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classification of higher dimensional varieties. Therefore, it is important to understand the geometry of the image of the Iitaka fibration.

In this note, we characterize the irregularity of Y_L in terms of dimension of certain cohomological support loci. To be more precise, let

$$V_m(L) = V_m := \{P \in \text{Pic}^0(X) \mid h^0(X, L^{\otimes m} \otimes P) \neq 0\},$$

and $V = \bigcup_{m \geq 1} V_m$. We show that the maximal irreducible component of V passing through the origin, denoted by G , is in fact the subgroup $f^* \text{Pic}^0(Y_L) \subset \text{Pic}^0(X)$. In particular, $q(Y_L) = \dim G$.

In general, the locus V need not be a group, it could have infinitely many irreducible components and there may exist $P, Q \in \text{Pic}^0(X)$ such that $f_{L \otimes P}$ and $f_{L \otimes Q}$ are not birational. Nevertheless, we show that when one considers the case of the canonical divisor $L = K_X$, then the set $V(K_X)$ is a subgroup of $\text{Pic}^0(X)$ such that its components consist of finitely many torsion translates of $\text{Pic}^0(Y)$. Moreover, we show that, for all $P \in V$, one has that the Iitaka fibration $f_{K_X \otimes P}$ is birational to f_{K_X} .

Finally, we study the case when X is of maximal Albanese dimension. Chen and Hacon (2001) have shown that the translates through the origin of the components of $V_1(K_X)$ generate $\text{Pic}^0(X)$. In Sec. 4 we show that the locus T given by the intersection of all (translates through the origin of the components of $V_1(K_X)$) plays a fundamental role in understanding the geometry of X . In particular we show that: *Let X be of general type and maximal Albanese dimension. If $P_1(X) = 1$ then $q(X) = \dim(X)$.* Moreover, we construct examples of such varieties. This answers a question of Koll  r (1995, Conjecture (17.9.2)).

1.1. Notation and Conventions

- (1.1) Throughout this article, we work over the field of complex numbers $\mathbb{C}$. X will always denote a projective variety.
- (1.2) $|D|$ will denote the linear series associated to the divisor D . We do not distinguish between line bundles, linear equivalence classes of divisors and invertible sheaves. Let $|V| \subset |L|$ be a linear subsystem. A log resolution of $|V|$ is a proper birational morphism $\mu: X' \rightarrow X$ such that X' is smooth, $\mu^*|V| = |W| + F$, where $|W|$ is base point free and the union of support of F and the exceptional set of μ is a divisor with normal crossings support.
- (1.3) For a real number a , let $\lfloor a \rfloor$ be the largest integer $\leq a$ and $\lceil a \rceil$ be the smallest integer $\geq a$. For a $\mathbb{Q}$ -divisor $D = \sum a_i D_i$, let $\lfloor D \rfloor = \sum \lfloor a_i \rfloor D_i$ and $\lceil D \rceil = \sum \lceil a_i \rceil D_i$.
- (1.4) We will denote by $A(X)$ the Albanese variety of X , by $\text{alb}_X: X \rightarrow A(X)$ the Albanese morphism. As usual $\text{Pic}^0(X)$ is the abelian variety dual to $A(X)$ parameterizing all topologically trivial line bundles on X . $\text{Pic}^0(X)_{\text{tors}}$ will denote the set of torsion elements in $\text{Pic}^0(X)$. We will denote by $m: A \times A \rightarrow A$ the group law of an abelian variety A . Given subsets $Z, W \subset A$, we define $Z + W := \text{image}(m|_{Z \times W}: Z \times W \rightarrow A)$.
- (1.5) Let $\mathcal{F}$ be a coherent sheaf on X , then $h^i(X, \mathcal{F})$ denotes the complex dimension of $H^i(X, \mathcal{F})$. In particular, the plurigenera $h^0(X, \omega_X^{\otimes m})$ are denoted by $P_m(X)$ and the irregularity $h^0(X, \Omega_X^1)$ is denoted by $q(X)$.

(1.6) Let L be a Cartier divisor on X . If $h^0(X, L) > 0$, then there is a rational map $\phi_{|L|}: X \rightarrow \mathbb{P}(H^0(X, L))$ defined by the sections of L . The Iitaka dimension of a line bundle L is defined as

$$\kappa(L) := \max\{\dim \phi_{|mL|}(X); m \in \mathbb{N}\}.$$

If $|mL| = \emptyset$ for all $m > 0$, we set $\kappa(L) = -\infty$. A nonsingular representative of the Iitaka fibration of X is a morphism of smooth complex projective varieties $f'_L = f': X' \rightarrow Y$ such that for all sufficiently big and divisible integers m , $f': X' \rightarrow Y$ is birational to $f_{|mL|}: X \rightarrow f_{|mL|}(X)$. It is characterized up to birational equivalence by the following properties:

- (i) $f': X' \rightarrow Y$ is an algebraic fiber space (i.e. it is surjective with connected fibers).
- (ii) $\dim(Y) = \kappa(L)$.
- (iii) $\kappa(X'_y, L|_{X'_y}) = 0$ (where X'_y is a generic geometric fiber of f').

There is a semi-group $N(L) = N(X, L) := \{m \in \mathbb{N}; H^0(X, L^{\otimes m}) \neq 0\}$. The exponent $e(L)$ is the greatest common divisor of all elements of $N(L)$. For all $m \gg 0$, one has that $m \in N(L)$ iff $e(L) | m$.

2. COHOMOLOGICAL SUPPORT LOCI

Lemma 2.1. *Let X be a closed subvariety (reduced and irreducible) of an abelian variety A passing through the origin. If X is closed under the group law, then X is an abelian subvariety.*

Proof. Let g be the dimension of X . Assume that $g > 0$. Since X is closed under multiplication, we have an induced morphism

$$m: X \times X \rightarrow X.$$

Let F_x be the fiber over $x \in X$. It is clear that $\dim F_x \geq g$. On the other hand, one notices that $F_x \cap (\{a\} \times X)$ consists of at most one point for all $a \in X$. This, together with the fact that $\dim F_x \geq g$, shows that $F_x \cap (\{a\} \times X)$ has exactly one point for all $a \in X$. Let $x = 0$ then it follows that X is a subgroup of A hence an abelian subvariety. $\square$

Let X be a smooth projective variety with $q(X) > 0$, L a Cartier divisor on X with $\kappa(L) \geq 0$ and exponent $e := e(L)$. We define

$$V_m = V_m(X, L) := \{P \in \text{Pic}^0(X) \mid h^0(X, L^{\otimes m} \otimes P) \neq 0\}.$$

By semi-continuity, each component of V_m is a closed subset of $\text{Pic}^0(X)$. Let $V = \bigcup_{m=1}^{\infty} V_m$. Then $V \subset \text{Pic}^0(X)$ is a semi-group. We define G_m to be the union of all irreducible components of V_m passing through the origin and let $G = \bigcup_{m>0} G_m$.

Note that G_m is non-empty only when $m \in N(L)$. Recall that for all $m \gg 0$, we have $m \in N(L)$ iff $e(L)|m$.

Lemma 2.2. *There exists an integer $m > 0$ such that $G = G_m$ is an abelian subvariety of $\text{Pic}^0(X)$.*

Proof. Pick any maximal irreducible component $W \subset G$, that is, if W' is an irreducible component of G containing W then $W = W'$. (This is possible since $\text{Pic}^0(X)$ is of course Noetherian). Assume that $W \subset G_{m_0}$. We claim that $W = G$ and W is closed under multiplication. Then we are done by Lemma 2.1. To see the claim, observe that if Z is any irreducible component of G_m , then $Z + W$ is irreducible since it is the image of $m : Z \times W \rightarrow \text{Pic}^0(X)$. Moreover,

$$Z + W \subset G_{m_0} + G_m \subset G_{m_0+m} \subset G.$$

Let W' be an irreducible component of G_{m_0+m} containing $Z + W$. It follows that $W \subset Z + W \subset W'$. By the maximality of W , one has $W = W'$. In particular, $Z \subset W$ and hence $W = G$.

It then suffices to check that W is closed under multiplication. But $W \subset W + W \subset G + G = G = W$ and hence $W + W = W$. $\square$

Corollary 2.3. *There exists an integer $t_0 > 0$ such that $G_{te} = G$ for all $t \geq t_0$.*

Question. Are the loci V (resp. V_m) union of translates of subgroups of $\text{Pic}^0(X)$?

Remark. It is easy to see that if $k(L) = 0$ and $P \in V$ is a non torsion element, then $P^\vee$ is not in V . In particular, $\dim G = 0$. The following example, shows that $\dim V_m$ could be positive even if $\kappa(L) = 0$.

Example 2.4. Consider $A = E \times E$ a product of elliptic curves and let $M = p_1^*H + p_2^*P$ with H ample of degree 1 and $P \in \text{Pic}^0(E) - \text{Pic}^0(E)_{\text{tors}}$. Let $X = \mathbb{P}(\mathcal{O}_A \oplus M)$, $\pi : X \rightarrow A$ and $L := \mathcal{O}_X(1)$. Since $\pi : X \rightarrow A$ is a projective bundle, it follows that π is the Albanese morphism of X . Since

$$\pi_*\mathcal{O}_X(m) = S^m(M \oplus \mathcal{O}_A) = M^{\otimes m} \oplus M^{\otimes m-1} \oplus \cdots \oplus M \oplus \mathcal{O}_A,$$

one sees that

$$V_m = \{\mathcal{O}_A\} \cup \bigcup_{i=1}^m (p_1^*\text{Pic}^0(E) + ip_2^*P^\vee).$$

We have

- (1) $\kappa(L) = 0$ and $\dim G(L) = 0$.
- (2) For all $m > 0$, $Q \in (p_1^*\text{Pic}^0(E) + mp_2^*P^\vee)$, $\kappa(L \otimes Q) = 1$ and $\dim V_m = \dim V = 1$.

This gives an example in which $\kappa(L) = 0$ but $\kappa(L \otimes P) > 0$ for some $P \in \text{Pic}^0(X)$. Moreover, $V = \bigcup V_m$ is a semi-group (with infinitely many components) but not a group.

Consider the following diagram relating the Iitaka fibration f_L and the Albanese morphism alb_X

$$\begin{array}{ccc} X & \xrightarrow{\text{alb}_X} & A(X) \\ f_L \downarrow & & \psi_L \downarrow \\ Y_L & \xrightarrow{\text{alb}_Y} & A(Y_L). \end{array}$$

Notice that f_L is not a morphism but simply a rational map. However there exists a birational model $\mu: X' \rightarrow X$ such that the induced map $f'_L: X' \rightarrow Y_L$ is a morphism. Let $\pi_L: X' \rightarrow A(Y_L)$ be the induced map.

Lemma 2.5. *If $P \in V_m$ then for all sufficiently big and divisible integers, $s > 0$ one has that $sP + \text{Pic}^0(Y_{L^{\otimes m \otimes P}}) \subset V$. In particular, if $\kappa(L) \geq 0$, then $\text{Pic}^0(Y_L) \subset G(L)$.*

Proof. Let $N = L^{\otimes m} \otimes P$. Fix H an ample line bundle on $A(Y_{L^{\otimes m \otimes P}})$. For $k \gg 0$ sufficiently divisible, we have that $h^0(X', \mu^* N^{\otimes k} \otimes \pi^* H^\vee) > 0$. Clearly, $\mu_* \mathcal{O}_{X'} = \mathcal{O}_X$ and $R^i \mu_* \mathcal{O}_{X'} = 0$ for all $i > 0$. Since $\mu^* N^{\otimes k} \otimes \pi^* H^\vee = \mu^*(N^{\otimes k} \otimes \text{alb}_X^* \psi_N^* H^\vee)$, by the projection formula, we have $h^0(X, N^{\otimes k} \otimes \text{alb}_X^* \psi_N^* H^\vee) > 0$. It is easy to see that for any integer $j \gg 0$, one has $h^0(H^j \otimes Q) > 0$ for any $Q \in \text{Pic}^0(Y_{L^{\otimes m \otimes P}})$. It follows that $h^0(X, N^{\otimes kj} \otimes Q^{\otimes j}) > 0$, i.e., $kjP + jQ \in V_{mkj}$. The Lemma follows by setting $s = kj$.

In particular, take $P = \mathcal{O}_X \in V_m$, then $\mathcal{O}_X \in \text{Pic}^0(Y_L) \subset V_{mkj}$ and hence $\text{Pic}^0(Y_L) \subset G$. $\square$

It follows that for any $m > 0$ and $P \in \text{Pic}^0(X)$, we have $q(Y_{L^{\otimes P}}) \leq \dim G(L \otimes P) \leq \dim V(L)$. Recall though that it is possible $\dim V(L) > G(L)$.

It also follows that if L is big, then there exists an integer $m_0 > 0$ such that $V_m = \text{Pic}^0(X)$ for all $m \geq m_0$.

Lemma 2.6. (i) $\psi: A(X) \rightarrow A(Y)$ is surjective and has connected fibers.

(ii) Let $K := \ker(\psi)$. Then there is an exact sequence

$$0 \rightarrow \text{Pic}^0(Y) \rightarrow \text{Pic}^0(X) \rightarrow \text{Pic}^0(K) \rightarrow 0.$$

(iii) Fix a general fiber $i: X_y \rightarrow X$. Let

$$H := \ker(i^*: \text{Pic}^0(X) \rightarrow \text{Pic}^0(X_y)).$$

Then, $H \supset \text{Pic}^0(Y)$ and $H/\text{Pic}^0(Y)$ is a finite group.

Proof. (i) See Hacon and Pardini (to appear) Proposition 2.1.b.

(ii) This is an immediate consequence of i) obtained by dualizing the exact sequence of abelian varieties

$$0 \rightarrow K \rightarrow A(X) \rightarrow A(Y) \rightarrow 0.$$

(iii) Let J_y be the abelian subvariety generated by the translates of $\text{alb}_X(X_y)$ through origin. Since there are at most countably many sub-abelian varieties of a fixed abelian variety, we have that $J = J_y$ does not depend on $y \in Y$ for general y . Let $B := A(X)/J$. Then $Y \rightarrow A(Y)$ factors through B , and hence there is a map of abelian varieties $A(Y) \rightarrow B$ inducing the map $Y \rightarrow B$. The image of X in $A(X)$ generates $A(X)$, hence its image in B generates B as well. It follows that image of $A(Y)$ in B generates B . And hence, one sees that $B = A(Y)$ i.e. $K = J$.

We have the following commutative diagram

$$\begin{array}{ccc} \text{Pic}^0(X) & \longrightarrow & \text{Pic}^0(K) \\ i^* \downarrow & & \searrow \\ \text{Pic}^0(X_y) & \xleftarrow{\alpha} & \text{Pic}^0(J), \end{array}$$

where $i: X_y \rightarrow X$ is the inclusion and α is induced from $\text{alb}_X: X_y \rightarrow J$. Note that, since $A(X_y) \rightarrow J$ is surjective, the homomorphism α has finite kernel. The assertion now follows from ii). $\square$

Theorem 2.7. $G(L) = \text{Pic}^0(Y_L)$.

Proof. By Lemma 2.5, one has $G(L) \supset \text{Pic}^0(Y_L)$.

If $\kappa(L) = 0$, it suffices to prove that $\dim G = 0$. This is clear as otherwise one may consider $m > 0$ such that $G(L) \subset V_m$ and the morphisms

$$|mL + P| \times |mL - P| \rightarrow |2mL|$$

for $P \in G$. It is easy to see that if $\dim G > 0$ then $\dim |2mL| > 0$ a contradiction.

Assume now that $\kappa(L) > 0$. If $P \in G(L)$, then for some $m \gg 0$, $h^0(L^{\otimes m} \otimes P) \neq 0$ in particular $h^0(X_y, L^{\otimes m} \otimes P|_{X_y}) \neq 0$. It follows that $i^*G(X, L) \subset V_m(X_y, L|_{X_y})$. And hence $i^*G(X, L) \subset G(X_y, L|_{X_y})$. However, $G(X_y, L|_{X_y}) = \{\mathcal{O}_{X_y}\}$ as we have seen in the preceding paragraph. Thus we have

$$i^*G(X, L) = G(X_y, L|_{X_y}) = \{\mathcal{O}_{X_y}\}.$$

By Lemma 2.6, $G(X, L) \subset H$. Since $G(X, L)$ is connected and contains the origin, it follows that $G(X, L) \subset \text{Pic}^0(Y_L)$. $\square$

3. IITAKA FIBRATIONS OF CANONICAL DIVISORS

From now on we will assume $L = K_X$ and $\kappa(X) \geq 0$.

Proposition 3.1. Fix an integer $m \geq 2$. For all $Q \in \text{Pic}^0(X)_{\text{tors}}$ and all $P \in \text{Pic}^0(Y_{K_X})$ one has

$$h^0(\omega_X^{\otimes m} \otimes P \otimes Q) = h^0(\omega_X^{\otimes m} \otimes Q).$$

Proof. Just follow the proof of Hacon and Pardini (to appear) Proposition 2.12. $\square$

Theorem 3.2. *The loci $V_m(K_X)$ consist of a finite union of torsion translates of abelian subvarieties of $\text{Pic}^0(X)$.*

Proof. By a result of Simpson, the loci $V_1(K_X)$ are torsion translates of abelian subvarieties of $\text{Pic}^0(X)$. Fix $m \geq 2$, and let $P \in V_m(K_X)$. Let $\mu: X' \rightarrow X$ be a log resolution of the non empty linear series $|r((m-1)K_X + P)|$. Let D be a general member of $\mu^*|r((m-1)K_X + P)|$. D has normal crossings support. Let $N := \mathcal{O}_{X'}((m-1)K_{X'} + P - \lfloor \frac{D}{r} \rfloor)$, then the locus

$$V_1(\omega_{X'} \otimes N) := \{Q \in \text{Pic}^0(X) \mid h^0(\omega_{X'} \otimes N \otimes Q) > 0\}$$

consist of a finite union of torsion translates of subtori of $\text{Pic}^0(X)$ (cf. Clemens and Hacon, to appear, Sec. 7; Simpson, 1993). Comparing base loci, it is easy to see that $h^0(\omega_{X'} \otimes N) = h^0(\omega_{X'}^{\otimes m} \otimes P)$ and for all $Q \in T_P$ we have

$$h^0(\omega_{X'}^{\otimes m} \otimes Q) = h^0\left(\omega_{X'}\left(\left\lfloor \frac{D}{r} \right\rfloor\right) \otimes N \otimes Q \otimes P^\vee\right) \geq h^0(\omega_{X'} \otimes N \otimes Q \otimes P^\vee) > 0.$$

Hence for each $P \in V_m(K_X)$ there exists T_P a torsion translate of a subtorus such that $P \in T_P \subset V_m(K_X) \subset \text{Pic}^0(X)$. It follows that $V_m(K_X) = \bigcup T_P$. Since $V_m(K_X) \subset \text{Pic}^0(X)$ is closed, it follows that $V_m(K_X)$ consists of a finite union of the T_P . $\square$

Lemma 3.3. *If $\kappa(X) = 0$ then there exists an integer $c(X) = c > 0$ and an element $P \in \text{Pic}^0(X)_{\text{tors}}$ such that for all $m > 0$, the set $V_m(K_X)$ is non-empty if and only if c divides m . Moreover, if c divides m , then $V_m = \{P^{\otimes \frac{m}{c}}\}$. In particular V is a subgroup of $\text{Pic}^0(X)$.*

Proof. It is easy to see that since $\kappa(X) = 0$, each V_m can have at most 1 torsion point. Therefore $V_m(K_X)$ consists of at most one element say $P_m \in \text{Pic}^0(X)_{\text{tors}}$. Let

$$c := \min\{m > 0 \mid V_m \neq \emptyset\}.$$

It is clear that $V_m \neq \emptyset$ for all m divisible by c . Let d be any positive integer such that $V_d \neq \emptyset$. We claim that $c \mid d$. We may write $d = cq + r$ with $q > 0$ and $0 \leq r \leq c-1$. Let D_c (resp. D_d) be the unique divisor in the linear series $|cK_X + P_c|$ (resp. $|dK_X + P_d|$). Since V_{cd} has only one point, we have $dP_c = cP_d$ and moreover $cD_d = dD_c$ as $h^0(cdK_X + dP_c) = 1$.

Consider now the divisor $D_d - qD_c$. It is effective since $c(D_d - qD_c) = rD_c$ is effective. It follows that $h^0(rK_X + P_d - qP_c) \neq 0$ and hence $V_r \neq \emptyset$. One sees that $r = 0$. Therefore, $V_m(K_X)$ is non-empty iff c divides m . The Lemma now follows easily. $\square$

Corollary 3.4. *Let $\tilde{H} := i^{*-1}(V(X_Y))$. For all $m > 0$ one has $V_m \subset \tilde{H}$. Moreover, $V \subset \tilde{H}$ is a subgroup of finite index containing G .*

Proof. For any $Q \in V_m$, if $h^0(\omega_X^{\otimes m} \otimes Q) \neq 0$, then $h^0(\omega_{X_Y}^{\otimes m} \otimes Q) \neq 0$. Therefore, $V_m \subset \tilde{H}$ for all $m > 0$ and hence, $V \subset \tilde{H}$.

Since $\tilde{H}/G$ is finite and $G \subset V \subset \tilde{H}$, it suffices to show that the semigroup V is in fact a group.

To this end, pick any $P \in V$, we may assume that $P \in V_m$ for $m \geq 2$ since $V_1 \subset V_{1+t}$ for all sufficiently big and divisible t . By Proposition 3.1 and Theorem 3.2, one sees that P is in an irreducible component of the type $G + P_0$ with P_0 torsion. Let k be the order of P_0 . One sees that

$$-P \subset G - P_0 = (k-1)(G + P_0) \subset V.$$

It follows that V is a group. $\square$

Corollary 3.5.

$$q(X) - q(Y) \leq q(X_y).$$

Proof. Since $\kappa(X_y) = 0$, by Kawamata (1981), the map $\text{alb}_{X_y}: X_y \rightarrow A(X_y)$ is surjective. The map $X_y \rightarrow J$ is factored through $A(X_y)$. It follows that $A(X_y) \rightarrow J$ is surjective by the definition of J . Thus $q(X_y) = \dim A(X_y) \geq \dim J$. But $J = K$ and $\dim K = \dim A(X) - \dim A(Y)$. The inequality follows. $\square$

Corollary 3.6. *X is of maximal Albanese dimension if and only if Y is of maximal Albanese dimension and*

$$q(X) - q(Y) = \dim X - \dim Y.$$

Proof. If X is of maximal Albanese dimension, then $\text{alb}_X: X \rightarrow A(X)$ is generically finite. It follows that $\text{alb}_X: X_y \rightarrow A(X)$ is generically finite. Therefore, $X_y \rightarrow J$ is an étale map of abelian varieties and $\dim X_y = \dim J$. Moreover, since alb_X is generically finite, for a general point $z \in \text{alb}_Y(Y)$, the preimage in X consists of at most finite union of fibers X_y . Thus alb_Y is generically finite and hence Y is of maximal Albanese dimension.

On the other hand, let $Y \rightarrow S \rightarrow A(Y)$ be the Stein factorization. And let $T := S \times_{A(Y)} A(X)$. It suffices to check that $X \rightarrow T$ is generically finite. Notice that $\kappa(X_y) = 0$ and $q(X_y) \geq \dim X_y$. It follows by Kawamata (1981) that X_y is birational to an abelian variety and hence that $X_y \rightarrow K$ is generically finite. Therefore $X_y \rightarrow T$ is generically finite, and so is $X \rightarrow T$. $\square$

Corollary 3.7. *For all $P \in \text{Pic}^0(Y)$, the Iitaka fibration $Y(K_X + P)$ is birational to the Iitaka fibration $Y(K_X)$.*

Proof. Let $P, Q \in \text{Pic}^0(Y)$ such that $P = 2Q$. From the morphism

$$|mK_X| \times |m(K_X + P)| \rightarrow |2m(K_X + Q)|,$$

one sees that $Y(K_X + Q)$ dominates $Y(K_X)$ and $Y(K_X + P)$. By Proposition 3.1,

$$\dim Y(K_X + Q) = \dim Y(K_X + P) = \dim Y(K_X).$$

since the various Iitaka fibrations $X \rightarrow Y(\dots)$ have connected fibers, then $Y(K_X + Q)$ is birational to $Y(K_X)$ and to $Y(K_X + P)$. $\square$

Lemma 3.8. *Let $a : X \rightarrow A(X)$ be the Albanese map. If $q(Y) = 0$, then $a_*\omega_X$ is a homogeneous vector bundle. In particular, $c_1(a_*\omega_X) = 0$.*

Proof. Let $a = \text{alb}_X$. If $q(Y) = 0$, then V_1 is supported on finitely many points. We recall that by definition

$$V^i(a_*\omega_X) := \{P \in \text{Pic}^0(A(X)) \mid h^i(a_*\omega_X \otimes P) \neq 0\}.$$

By Chen and Hacon (to appear) Proposition 2.2, we have

$$V_1 = V^0 \supseteq V^1 \supseteq \dots \supseteq V^q,$$

where $q = q(X)$. It follows that all the V^i are supported on finitely many points.

We now follow Lazarsfeld (Personal communication) to show that $a_*\omega_X$ is a homogeneous vector bundle. Let p_i be the projection of $A(X) \times \text{Pic}(X) = A \times \hat{A}$ on to the i th factor and $\mathcal{P}$ be the normalized Poincaré line bundle on $A \times \hat{A}$. Then $RSa_*\omega_X = Rp_{2*}(p_1^*a_*\omega_X \otimes \mathcal{P})$ is the Fourier-Mukai transform of $a_*\omega_X$. By Mukai (1981), it is enough to show that $R^gSa_*\omega_X \cong RSa_*\omega_X$ is a coherent sheaf supported on finitely many points. To see this, recall that for an open affine $U = \text{Spec}(B) \subset \hat{A}$, there is a complex of finitely generated free B -modules, denoted by $E^\bullet$, such that

$$R^iSa_*\omega_X \cong H^i(E^\bullet),$$

and

$$H^i(X_y, a_*\omega_X \otimes P_y) \cong H^i(E^\bullet \otimes k(y)).$$

It suffices to show that $H^i(E^\bullet) = 0$ for all $i < q$ and $H^q(E^\bullet)$ is supported on finite point. We may assume that B is local since localization is flat. Let

$$W^i(E^\bullet) := \text{coker}(E^{i-1} \rightarrow E^i).$$

It is easy to see that $W^i(E^\bullet) \otimes k(y) = W^i(E^\bullet \otimes k(y))$ since $\otimes$ is right exact. One has exact sequence

$$0 \rightarrow H^i(E^\bullet) \rightarrow W^i(E^\bullet) \rightarrow E^{i+1}.$$

Our hypothesis on the cohomologies of $a_*\omega_X$ implies that $H^i(E^\bullet)$ is Artinian for all i . Thus by Mumford's acyclic lemma (Mumford, p. 127), we are done. $\square$

4. VARIETIES OF MAXIMAL ALBANESE DIMENSION

Throughout this section, we assume that X is of maximal Albanese dimension i.e., $\dim(\text{alb}_X(X)) = \dim X$.

In (17.9.3) Kollár (1995) Conjectured that *If X is of general type and Maximal Albanese dimension, then $\chi(X, \mathcal{O}_X) > 0$.* A consequence of this conjecture would be:

- (i) $V_1(X, K_X) = \text{Pic}^0(X)$ and
- (ii) (cf. (17.9.2) of Kollár, 1995) that $P_1(X) \geq 2$.

In Ein and Lazarsfeld (1997) show that $\chi(X, \mathcal{O}_X) \geq 0$, and they produce a counter-example to the above conjecture of Koll  r (1995, 17.9.3). Chen and Hacon (2001), show that:

Theorem 4.1. *If X is of maximal Albanese dimension then (the translates through the origin of) the irreducible components of $V_1(X, K_X)$ generate $\text{Pic}^0(Y)$, and for all $m \geq 2$ one has $G_m(X, K_X) = \text{Pic}^0(Y)$.*

In particular if X is of general type, they generate $\text{Pic}^0(X)$ (see (i) above).

Let X be of maximal Albanese dimension. If $\chi(\mathcal{O}_X) = 0$, let S'_i be the translates of the components of $V_1(X, K_X)$ through the origin and $T := \cap S'_i$.

Theorem 4.2. *Let X be a variety of general type, maximal Albanese dimension with $\chi(\omega_X) = 0$. Let $A(X) \rightarrow T^\vee$ be the corresponding map of abelian varieties, and $\pi: X \rightarrow T^\vee$ the induced morphism. For general w in $W := \pi(X)$, one has that $\chi(\omega_{X_w}) = 0$ and $\text{alb}_X(X_w)$ is a translate of the abelian subvariety $\ker(A(X) \rightarrow T^\vee)$. In particular, $\dim T \neq 1$ and if $\dim T = 0$ then $q(X) = \dim(X)$.*

Proof. Recall that Ein and Lazarsfeld (1997, Theorem 3), the Albanese image $Z := \text{alb}_X(X) \subset \text{Alb}(X)$ is fibered by tori. In fact Ein and Lazarsfeld (1997) show the following: Let S be any positive dimensional component of $V^0(\omega_X) = V_1(X, K_X)$. Consider the quotient map $\pi: A := \text{Alb}(X) \rightarrow C$ dual to the inclusion $S \hookrightarrow \text{Pic}^0(A)$. Let $g: X \rightarrow C$ be the induced morphism, then $\dim g(X) = \dim(X) - (\dim(A) - \dim(C))$, i.e., $Z \rightarrow g(X)$ is fibered by tori. By Chen and Hacon (2001, Theorem 2.3), the translates through the origin of the components of $V_1(\omega_X)$ generate $\text{Pic}^0(X)$. X is of general type and hence not ruled by tori. Assume that the intersection of the translates through the origin of the components of $V_1(\omega_X)$ is just $\{\mathcal{O}_X\}$. Let S_i be any component of $V_1(\omega_X)$ and B_i be the kernel of the corresponding projection of abelian varieties $\pi_i: A \rightarrow C_i := S_i^\vee$. Then the union of the B_i generates A . As observed above, by the result of Ein and Lazarsfeld (1997), $Z + B_i = Z$ and hence $Z = A$, i.e., $q(X) = \dim X$.

Suppose now that $\dim T > 0$ and consider the following dual exact sequences of abelian varieties

$$\begin{aligned} 0 \longrightarrow T \longrightarrow \text{Pic}^0(X) \xrightarrow{\phi} \text{Pic}^0(X)/T \longrightarrow 0, \\ 0 \longrightarrow F \longrightarrow A(X) \longrightarrow T^\vee \longrightarrow 0. \end{aligned}$$

Let $\bar{S}_i := S_i/T \subset \text{Pic}^0(F) = \text{Pic}^0(X)/T$ and $\pi: X \rightarrow T^\vee$ the induced morphism. For any $P \in \text{Pic}^0(X)_{\text{tors}}$ such that $P \notin V_1(X, K_X)$, one has that

$$\phi^{-1}(\phi(P)) = P + T \cap V_1(X, K_X) = \emptyset.$$

So for all $Q \in T$ one has $h^0(X, \omega_X \otimes P \otimes \pi^*Q) = 0$ and therefore also $h^i(X, \omega_X \otimes P \otimes \pi^*Q) = 0$ for all $i \geq 0$. By a result of Koll  r (1987, Corollary 3.3) one sees that for all $Q \in T$, the group $H^i(\pi_*(\omega_X \otimes P) \otimes Q)$ is a direct summand of the group $H^i(X, \omega_X \otimes P \otimes \pi^*Q)$ and hence also vanishes. By a result of Mukai (1981) it follows that $\pi_*(\omega_X \otimes P) = 0$. So for general $w \in W = \pi(X)$, one has $h^0(X_w, \omega_{X_w} \otimes P) = 0$.

Therefore, $\chi(\omega_{X_w}) = 0$ and the locus

$$V_1(X_w, K_{X_w}, \text{Pic}^0(F)) := \{R \in \text{Pic}^0(F) \mid h^0(\omega_{X_w} \otimes R) > 0\}$$

(which is determined by its torsion points) is a subset of $V_1(X, K_X)/T$. It follows that the intersection of the translates through the origin of the components of $V_1(X_w, K_{X_w}, \text{Pic}^0(F))$ is just $\{\mathcal{O}_F\}$. Therefore, arguing as in the preceding paragraph, one sees that $\text{alb}_X(X_w) = F$ is an abelian subvariety of $A(X)$. $\square$

Corollary 4.3. *Let X be a 3-fold of general type and maximal Albanese dimension. If $\chi(\omega_X) = 0$ then $q(X) = \dim X$.*

Proof. We may assume that $\dim X > \dim T > 0$ and hence X_w is of general type and has dimension 1 or 2. By the classification theory of curves and surfaces $\chi(\omega_{X_w}) > 0$. This is the required contradiction. $\square$

Corollary 4.4. *Let X be a variety of general type, maximal Albanese dimension. If $P_1(X) = 1$ then $q(X) = \dim X$.*

Proof. If $P_1(X) = 1$ then $\chi(\omega_X) = 0$. Let $\tilde{W} \rightarrow W$ be an appropriate desingularization (of $W = \pi(X)$ as above). Since X is of maximal Albanese dimension, one has $P_1(\tilde{W}) = 1$ and hence since W is a subvariety of the abelian variety $T^\vee$, it follows that $W = T^\vee$. Since $\text{alb}_X(X_w) = F$ one sees that $\text{alb}_X(X) = A(X)$ and hence $q(X) = \dim(X)$. $\square$

Corollary 4.5. *Let X be a variety of general type, maximal Albanese dimension with $\chi(\omega_X) = 0$. If $\dim T = 0$, then $h^1(X, P) = 0$ for all $P \in \text{Pic}^0(X) - \{\mathcal{O}_X\}$.*

Proof. Assume that $h^1(X, P) > 0$ for some $P \in \text{Pic}^0(X) - \{\mathcal{O}_X\}$. By Simpson (1993), we may assume that P is torsion. Consider the étale cover $Y \rightarrow X$ corresponding to the subgroup $\langle P \rangle \subset \text{Pic}^0(X)$. Clearly Y is of general type, maximal Albanese dimension and $\chi(\omega_Y) = 0$ so by the Theorem 4.2

$$\dim(Y) = q(Y) = h^1(\mathcal{O}_Y) = \sum h^1(X, P^{\otimes i}).$$

In particular $h^1(X, P^{\otimes i}) = 0$ for $i > 0$. $\square$

The following example shows that Kollàr's conjecture (17.9.2) of Kollàr (1995) also fails:

Example. Let $d: D \rightarrow E$ be the $\mathbb{Z}_2^2$ cover of an elliptic curve defined by $d_*\mathcal{O}_D = \mathcal{O}_E \oplus L^\vee \oplus P \oplus L^\vee \otimes P$ where $\deg(L) = 1$ and P is a non-zero 2-torsion element of $\text{Pic}^0(E)$. Then $g(D) = 3$ and we will denote by l, p, lp the elements of $\mathbb{Z}_2^2$ whose eigensheaves with eigenvalue 1 are $\mathcal{O}_E$ and $L^\vee, P, L^\vee \otimes P$ respectively. It follows that

$$\delta: D \times D \times D \rightarrow E \times E \times E$$

is a $\mathbb{Z}_2^6$ cover. Let $X := D \times D \times D/G$ where $G \cong \mathbb{Z}_2^4$ is the subgroup of $\mathbb{Z}_2^6$ generated by $\{(id, p, l), (p, l, id), (l, id, p), (p, p, p)\}$. By a direct check, one can see that

$$\begin{aligned} & (\delta_* \mathcal{O}_{D \times D \times D})^G \\ & \cong \mathcal{O}_E \boxtimes \mathcal{O}_E \boxtimes \mathcal{O}_E \oplus L^\vee \boxtimes L^\vee \otimes P \boxtimes P \oplus P \boxtimes L^\vee \boxtimes L^\vee \otimes P \oplus L^\vee \otimes P \boxtimes P \boxtimes L^\vee. \end{aligned}$$

The singularities of X are of type $\frac{1}{2}(1, 1, 1)$ i.e. locally isomorphic to $\mathbb{C}^3/\langle(-1, -1, -1)\rangle$. In particular all singularities are rational. Let $\tilde{X} \rightarrow X$ be a resolution of the singularities of X and $\nu: \tilde{X} \rightarrow E \times E \times E$ be the induced morphism. One can compute that

$$\nu_*(\omega_{\tilde{X}}) = \mathcal{O}_E \boxtimes \mathcal{O}_E \boxtimes \mathcal{O}_E \oplus L \boxtimes L \otimes P \boxtimes P \oplus P \boxtimes L \boxtimes L \otimes P \oplus L \otimes P \boxtimes P \boxtimes L.$$

Therefore, $\tilde{X}$ is a threefold of maximal Albanese dimension with $P_1(X) = 1$ and $q(X) = 3$.

We remark that this example easily generalizes to $\dim X \geq 3$. The product $X \times C$ with C a curve of genus 2, is a variety of general type and maximal Albanese dimension with $\chi(\omega_{X \times C}) = 0$ and $5 = q(X) > \dim X = 4$. Moreover, for any étale cover $\tilde{X} \rightarrow X$ one also has $\chi(\tilde{X}) = 0$. One can then construct (infinitely many) examples where $P_1(\tilde{X}) = P_1(X)$, $q(\tilde{X}) = q(X)$.

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**On the Irregularity of the Image of the
Iitaka Fibration[#]**

Jungkai A. Chen and Christopher D. Hacon

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